

LM Physics of Climate

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4 Moist convection: a two-component atmosphere

4.1 The two-component moist pseudo-adiabat

Now let us consider a non-condensable gas mixed with a condensable gas, e.g. a mixture of N_2 and H_2O at Earth conditions. As above, we will assume that condensation is efficient such that the partial pressure of the condensate is equal to its saturation vapour pressure, $P_c = P_{c,sat}$. We will additionally assume that any condensate is removed, i.e. that it rains out. Since this is an irreversible process, the resulting temperature profile is called the moist *pseudoadiabat* rather than an adiabat.

As with the dry adiabat, $dQ = 0$ for moist convection. However, Equation 2.22 ($dQ = M(c_p dT - \frac{dP}{\rho})$) requires a few extra terms to account for the different gas components and the effects of latent heat. Using the subscripts a (for ‘air’) and c to represent the properties of the non-condensable and condensable gases, respectively, we can rewrite Equation 2.22 as:

$$\begin{aligned} dQ &= M_a c_{p,a} dT + M_c c_{p,c} dT - \frac{M_a}{\rho_a} dP_a - \frac{M_c}{\rho_c} dP_c + L dM_c \\ &= M_a \left(c_{p,a} dT - \frac{1}{\rho_a} dP_a \right) + M_c \left(c_{p,c} dT - \frac{1}{\rho_c} dP_c \right) + L dM_c. \end{aligned} \quad (4.1)$$

where M_a and M_c are the total masses of each gas, P_a and P_c are their partial pressures, $\rho_a = M_a/V$ and $\rho_c = M_c/V$. Note that when condensation occurs, dM_c is negative (there is less condensable gas left), so for a positive latent heat, L , condensation results in a negative contribution to dQ , which ‘releases’ heat to the system.

We can rearrange Equation 4.1 by dividing out $M_c T$ and defining the mass mixing ratio, $r_c = M_c/M_a$:

$$\frac{dQ}{M_c T} = \frac{1}{r_c} \left(c_{p,a} \frac{dT}{T} - \frac{dP_a}{\rho_a T} \right) + \left(c_{p,c} \frac{dT}{T} - \frac{dP_c}{\rho_c T} \right) + \frac{L}{T} \frac{dM_c}{M_c}.$$

Next, we use the ideal gas law for each gas component to convert $1/(\rho T) = k_B/(\mu P)$ and multiply by r_c . The total mass of gas, $M = M_a + M_c$. Note that $dr_c = dM_c/M_a$ since M_a is constant (gas a does not condense):

$$(1 + r_c) \frac{dQ}{MT} = \left(c_{p,a} \frac{dT}{T} - \frac{k_B}{\mu_a} \frac{dP_a}{P_a} \right) + r_c \left(c_{p,c} \frac{dT}{T} - \frac{k_B}{\mu_c} \frac{dP_c}{P_c} \right) + \frac{L}{T} dr_c. \quad (4.2)$$

From here, we can express dQ in terms of $d \ln P_a$ and $d \ln T$ (the goal is ultimately to find $d \ln P_a / d \ln T$). Assuming that the parcel of gas remains saturated, we can state that $P_c = P_{sat}$ and

$$r_c = r_{sat} = \frac{M_{c,sat}}{M_a} = \frac{\rho_{c,sat}}{\rho_a} = \frac{\mu_c}{\mu_a} \frac{P_{sat}}{P_a}. \quad (4.3)$$

It is also useful to rearrange Equation 3.6 (a version of the Clausius-Clapeyron equation, $\frac{dP_{sat}}{dT} = \frac{L\mu}{k_B T^2} P_{sat}$) to give:

$$d \ln P_{sat} = \frac{L\mu}{k_B T^2} dT = \frac{L\mu}{k_B T} d \ln T, \quad (4.4)$$

and to express dr_{sat} as:

$$dr_{sat} = \frac{\mu_c}{\mu_a} d \left(\frac{P_{sat}}{P_a} \right) = \frac{\mu_c}{\mu_a} \frac{P_{sat}}{P_a} d \ln \left(\frac{P_{sat}}{P_a} \right) = r_{sat} (d \ln P_{sat} - d \ln P_a). \quad (4.5)$$

Substituting Equations 4.3, 4.4 and 4.5 into Equation 4.2, we get:

$$\begin{aligned} (1 + r_c) \frac{dQ}{MT} &= c_{p,a} d \ln T + r_{sat} c_{p,c} d \ln T - \frac{k_B}{\mu_a} d \ln P_a - r_{sat} \frac{k_B}{\mu_c} d \ln P_{sat} \\ &\quad + \frac{Lr_{sat}}{T} (d \ln P_{sat} - d \ln P_a) \\ &= (c_{p,a} + r_{sat} c_{p,c}) d \ln T - \left(\frac{k_B}{\mu_a} + \frac{Lr_{sat}}{T} \right) d \ln P_a \\ &\quad + \left(\frac{Lr_{sat}}{T} - r_{sat} \frac{k_B}{\mu_c} \right) d \ln P_{sat} \\ &= (c_{p,a} + r_{sat} c_{p,c}) d \ln T - \left(\frac{k_B}{\mu_a} + \frac{Lr_{sat}}{T} \right) d \ln P_a \\ &\quad + \left(\frac{L}{T} - \frac{k_B}{\mu_c} \right) \frac{L\mu_c}{k_B T} r_{sat} d \ln T \\ &= \left[c_{p,a} + r_{sat} \left(c_{p,c} + \frac{L}{T} \left(\frac{L\mu_c}{k_B T} - 1 \right) \right) \right] d \ln T \\ &\quad - \left[1 + \frac{L\mu_a}{k_B T} r_{sat} \right] \frac{k_B}{\mu_a} d \ln P_a \end{aligned}$$

Setting $dQ = 0$, we get

$$\frac{d \ln T}{d \ln P_a} = \frac{k_B}{\mu_a c_{p,a}} \cdot \frac{1 + \frac{L\mu_a}{k_B T} r_{sat}}{1 + r_{sat} \left(\frac{c_{p,c}}{c_{p,a}} + \frac{L}{c_{p,a} T} \left(\frac{L\mu_c}{k_B T} - 1 \right) \right)} \quad (4.6)$$

Note that in the limit of $r_{sat} \rightarrow 0$, this reduces to the dry adiabat:

$$\frac{d \ln T}{d \ln P_a} \rightarrow \frac{k_B}{\mu_a c_{p,a}} = \frac{c_{p,a} - c_{v,a}}{c_{p,a}} = \frac{\gamma - 1}{\gamma},$$

where we have used Equation 2.14 (Mayer's relation, $c_P - c_V = \frac{N_{mol} R}{M} = \frac{k_B}{\mu}$) to express k_B/μ_a in terms of the specific heat capacities.